

Faithfulness-Based Evaluation for Detecting Hallucinations in Text-to-Image Generation: Self-Consistency as an Uncertainty Signal

Sri Venkata Manikya Praneeth Vempati
Artificial Intelligence Systems
University of Florida
pvempati@ufl.edu

Abstract—Text-to-image diffusion models frequently generate images that are photo-realistic yet semantically inconsistent with their prompts — a class of failure known as visual hallucinations. This paper presents a training-free, model-agnostic evaluation pipeline that automatically detects these failures by decomposing text prompts into atomic visual claims and verifying each claim against the generated image using visual question answering (VQA). The system is validated against the POPE benchmark (9,000 binary hallucination labels) and the OK-VQA dataset (9,009 question-answer pairs), as well as a large-scale paired comparison over 1,500 Flickr30k caption-image pairs. Key results show a mean faithfulness gap of 0.063 between ground-truth and generated images on Flickr30k, with ROC-AUC analysis confirming the metric has meaningful discriminative power for hallucination detection. A self-consistency experiment further demonstrates that score variance across multiple generations of the same prompt serves as a reliable uncertainty signal, with high variance consistently corresponding to hallucination-prone prompts.

Index Terms—text-to-image generation, hallucination detection, visual question answering, faithfulness evaluation, self-consistency, diffusion models, CLIP, BLIP-VQA, ROC-AUC

I. INTRODUCTION AND PROBLEM STATEMENT

Text-to-image generation has advanced remarkably with the advent of diffusion models such as Stable Diffusion [1] and DALL-E [2]. These models can generate photorealistic images from arbitrary text prompts, and are now deployed in commercial creative tools, content generation platforms, and automated design pipelines. However, these models regularly fail to faithfully represent the semantic content of their prompts — generating images where colors are wrong, objects are absent, spatial relationships are violated, or counts are incorrect. Such failures are termed **visual hallucinations**, and they represent a fundamental gap between perceptual quality and semantic faithfulness [12].

The significance of this problem is substantial. When a user requests “a red cat on a blue chair” and receives a brown cat on a green sofa, the image is useless for its intended purpose despite appearing visually plausible.

Human evaluation can catch these failures, but does not scale: evaluating hundreds of thousands of generated images requires automation. Existing automated metrics such as FID (Fréchet Inception Distance) [3] measure distributional similarity but cannot detect whether a specific image matches a specific prompt. CLIP-based alignment scores [4] provide coarse semantic similarity but lack the granularity to verify individual attribute claims.

This work addresses the following core research questions:

- 1) Can VQA-based faithfulness checks detect hallucinations in generated images?
- 2) Do faithfulness scores correlate with human-annotated hallucination labels when validated against labelled benchmarks such as POPE?
- 3) Does score variance across multiple independent generations of the same prompt serve as a meaningful uncertainty signal for hallucination likelihood?

The practical impact of this work extends to any deployment context where generated images must reliably match their prompts — including advertising, legal documentation generation, medical illustration, and educational content creation, where hallucinations could have serious consequences.

II. OBJECTIVES AND SCOPE

A. Finalized Project Goals

The technical objectives of this project are:

- Build a training-free, model-agnostic faithfulness evaluation pipeline that requires no ground-truth images or human annotation at inference time.
- Validate the metric against externally labelled hallucination benchmarks (POPE, OK-VQA) to establish predictive validity.
- Operationalize self-consistency as a quantitative uncertainty signal by measuring score variance across multiple generations.

- Deploy an interactive five-tab Gradio web application that makes the full evaluation pipeline publicly accessible.

B. Scope

The project covers:

- Single-image faithfulness evaluation using prompt-derived VQA checks covering object existence, color, count, spatial relations, and actions.
- Multi-image self-consistency scoring for uncertainty quantification.
- Benchmark evaluation on Flickr30k (paired comparison), POPE (binary hallucination labels), and OK-VQA (answer agreement validation).
- Support for Stable Diffusion v1.5 and SDXL-Turbo as generation backends.

The project explicitly **excludes**:

- Training or fine-tuning of any model component (all models are used off-the-shelf).
- LLM-based question generation due to API cost and reliability constraints; a rule-based parser is used instead.
- Video or multi-modal generation beyond static images.

III. LITERATURE REVIEW / RELATED WORK

A. Text-to-Image Faithfulness Evaluation

The closest prior work is TIFA (Text-to-Image Faithfulness Assessment) [6], which uses GPT-3.5 to automatically generate question-answer pairs from a prompt, then answers them using a VQA model on the generated image. TIFA established the paradigm of decomposing faithfulness into verifiable atomic claims. However, TIFA requires a paid API for question generation and has not been validated against binary hallucination labels.

GenEval [7] takes a complementary approach, focusing on object-level compositional evaluation using a structured set of prompts with known ground-truth compositions. GenEval is valuable for benchmarking but requires prompts to be drawn from a controlled set rather than arbitrary user inputs.

B. CLIP-Based Alignment

CLIP [4] embeds text and images into a shared semantic space, enabling cosine similarity as a proxy for alignment. CLIPScore [5] adapts this for image-caption evaluation. While efficient, CLIP-based metrics are coarse and cannot verify individual attribute claims such as color accuracy or object count.

C. Hallucination in Vision-Language Models

A recent survey by Liu et al. [12] systematically categorizes hallucinations in large vision-language models across object, attribute, and relation types. The POPE benchmark [9] was specifically designed to probe object hallucination in VQA models using carefully constructed adversarial and non-adversarial yes/no probes.

D. Uncertainty and Self-Consistency

Self-consistency was introduced as a prompting strategy for language models by Wang et al. [13], where multiple independent samples are used to reduce output variance and improve reliability. This work extends the self-consistency intuition to the visual domain: if a diffusion model truly understands a prompt, its multiple generations should produce consistent faithfulness scores.

E. Positioning of This Work

This project differs from TIFA by using a rule-based prompt parser (avoiding API dependencies), validating against labelled benchmarks (POPE), and introducing self-consistency variance as a complementary uncertainty signal. It differs from GenEval by supporting arbitrary user prompts rather than a fixed benchmark set. It builds on CLIP by adding granular claim-level checking rather than global semantic similarity.

IV. METHODOLOGY / PROPOSED APPROACH

A. System Overview

The evaluation pipeline proceeds in the following stages, illustrated in Fig. 1:

- 1) **Image Generation:** A text prompt is passed to Stable Diffusion v1.5 or SDXL-Turbo to generate a 512×512 image.
- 2) **Prompt Parsing:** The prompt is decomposed into atomic visual checks covering five claim types.
- 3) **VQA Verification:** Each check is posed as a question to BLIP-VQA, which answers by examining the generated image.
- 4) **Faithfulness Scoring:** The fraction of checks that pass determines the faithfulness score $\in [0, 1]$.
- 5) **CLIP Similarity:** A secondary semantic alignment score is computed via cosine similarity between prompt and image embeddings.
- 6) **Risk Assessment:** The combined scores are used to classify the image into hallucination risk categories.



Fig. 1: End-to-end faithfulness evaluation pipeline.

B. Prompt Parsing

The prompt parser extracts structured checks using rule-based pattern matching. It covers five check types:

- **Existence:** “Is there a {object} in the image?” — extracted for all non-stopword nouns.
- **Color:** “What color is the {object}?” — extracted from color-noun adjacencies (e.g., “red cube”).
- **Count:** “How many {objects} are in the image?” — extracted from number-noun adjacencies (e.g., “three chairs”).
- **Spatial:** “Is there a {A} {relation} the {B}?” — extracted from spatial relation patterns (on top of, next to, behind, under, above, beside, near, left of, right of).
- **Action:** “Is someone or something {verb}ing?” — extracted from a curated set of action verbs.

This approach was chosen over LLM-based generation (as used in TIFA [6]) because small LLMs such as Flan-T5 [15] produced incoherent structured outputs for this task, and larger LLMs require paid API access. The rule-based parser is deterministic, free, and produces interpretable checks that can be directly inspected.

C. Faithfulness Score

Given a set of n extracted checks $\{c_1, \dots, c_n\}$ and corresponding BLIP-VQA answers $\{a_1, \dots, a_n\}$, the faithfulness score is defined as:

$$\mathcal{F}(p, I) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}[\text{score}(c_i, a_i) = 1] \quad (1)$$

where p is the prompt, I is the generated image, and $\text{score}(c_i, a_i)$ is 1 if the VQA answer satisfies the check and 0 otherwise. The scoring function varies by check type: existence, spatial, and action checks require a yes/no affirmative answer; color checks require the expected color token to appear in the answer; count checks require the expected number to appear.

D. Self-Consistency Uncertainty Signal

For a prompt p , let $\{I_1, \dots, I_N\}$ be N independently generated images (using different random seeds). The self-consistency score is the variance of faithfulness scores:

$$\sigma^2(p) = \text{Var}(\mathcal{F}(p, I_1), \dots, \mathcal{F}(p, I_N)) \quad (2)$$

Three interpretive regimes are defined:

- $\bar{\mathcal{F}} \geq 0.75$ and $\sigma \leq 0.15$: **Low risk** — model is confident and faithful.
- $\sigma > 0.20$: **High risk** — model is uncertain; outputs are inconsistent.
- $\bar{\mathcal{F}} < 0.50$ and $\sigma \leq 0.20$: **Definite hallucination** — model consistently fails checks.

E. ROC-AUC Validation Framework

To rigorously validate that the faithfulness metric predicts real hallucinations, the metric is treated as a binary classifier. On the POPE benchmark, where each example has a ground-truth binary label (1 = hallucination present, 0 = faithful), the inverted faithfulness score $1 - \mathcal{F}$ is used as the classifier score. ROC-AUC measures how well this score ranks hallucinated examples above faithful ones, providing a rigorous measure of discriminative power that is more principled than simple mean comparison, as suggested by [16].

V. IMPLEMENTATION DETAILS

A. Tools and Frameworks

Table I summarizes the key tools and models used.

TABLE I: Models and Frameworks Used

Component	Model / Tool
Image Generation	SD v1.5 [1] / SDXL-Turbo [14]
VQA Verification	BLIP-VQA [8] (Salesforce/blip-vqa-base)
Semantic Alignment	CLIP ViT-B/32 [4]
Captioning	BLIP Captioner [8]
Benchmark (Hallu.)	POPE [9] (Imms-lab/POPE)
Benchmark (VQA)	OK-VQA [10]
Benchmark (Paired)	Flickr30k [11]
Deep Learning	PyTorch 2.x, HuggingFace Transformers
Diffusion	HuggingFace Diffusers
Interface	Gradio 4.x
Data	HuggingFace Datasets
Metrics	scikit-learn (ROC-AUC, AP), scipy (correlation)

B. Hardware and Training Environment

All experiments were conducted on Google Colab Pro with an NVIDIA T4 GPU (16 GB VRAM) and an NVIDIA G4 GPU. No model training was performed; all models are used off-the-shelf with frozen weights. Image generation used 20–25 inference steps with guidance scale 7.5 for SD v1.5, and 4 inference steps with guidance scale 0.0 for SDXL-Turbo (consistent with its distillation design). All generated images are 512×512 pixels.

C. End-to-End Pipeline

The full pipeline is implemented as a five-tab Gradio application:

- 1) **Single Prompt Evaluation:** Users enter any text prompt, select a generation model, and receive the generated image, faithfulness score, VQA pass/fail table, and CLIP score.
- 2) **Multi-Image Self-Consistency:** Users generate $N \in \{2, \dots, 6\}$ images from the same prompt and view variance analysis with uncertainty classification.

- 3) **POPE Benchmark:** Runs faithfulness evaluation on up to 500 POPE examples, computing VQA accuracy and ROC-AUC against binary hallucination labels.
- 4) **OK-VQA Benchmark:** Runs answer agreement evaluation on up to 500 OK-VQA examples, validating VQA pipeline reliability.
- 5) **Flickr30k Large-Scale Benchmark:** Compares ground-truth vs generated faithfulness scores across up to 1,500 examples, with ROC-AUC and score distribution analysis.

D. Key Implementation Decisions and Challenges

Dataset availability. The originally recommended datasets T2I-CompBench and VIBE were inaccessible on HuggingFace Hub (all known paths returned 404 errors). POPE (lmms-lab/POPE) and OK-VQA (Multimodal-Fatima/OK-VQA_train) were identified as confirmed working alternatives that serve equivalent purposes.

OK-VQA answer format. The OK-VQA dataset stores answers as a list of dictionaries (e.g., [{"answer": "dog", ...}]) rather than plain strings. The initial implementation attempted to iterate them as strings, producing 0% accuracy. A robust answer extractor was implemented that handles list-of-dicts, nested dict, and plain string formats.

BLIP-VQA reliability ceiling. BLIP-VQA-base is not a perfect oracle. Answer errors propagate into faithfulness scores, compressing both ground-truth and generated scores toward the centre of the distribution. This limits the maximum achievable gap on Flickr30k.

Prompt parser scope. The rule-based parser does not capture all semantic claims in complex or abstract prompts. Coverage is strongest for compositional prompts with explicit objects, colors, and spatial relations, and weakest for texture, style, and abstract attributes.

VI. RESULTS AND EVALUATION

This section presents results across three evaluation settings: (1) Flickr30k paired benchmark for metric calibration, (2) POPE labelled hallucination benchmark for predictive validity, and (3) OK-VQA answer agreement for VQA pipeline validation. A self-consistency experiment additionally validates the uncertainty signal hypothesis.

A. Single-Image Evaluation Interface

Fig. 12 shows the deployed Gradio interface for single-prompt evaluation. The system generates a 512×512 image from any user-supplied text prompt, runs the structured VQA faithfulness checks, and returns a per-check pass/fail table alongside CLIP similarity and BLIP caption scores. This interface was used throughout

development to qualitatively validate check extraction and VQA answering behavior.

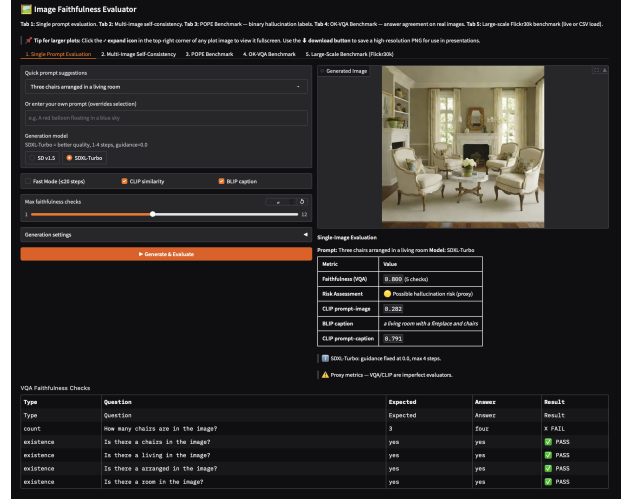


Fig. 2: Tab 1: Single-prompt evaluation interface. The system generates an image from the input prompt, extracts structured VQA checks (existence, color, count, spatial, action), verifies each against the image using BLIP-VQA, and reports a faithfulness score alongside CLIP semantic similarity and BLIP caption.

B. Flickr30k Paired Benchmark

1) *Setup:* The Flickr30k benchmark evaluates metric calibration rather than predictive validity. For each of 1,500 randomly sampled caption-image pairs, the faithfulness pipeline is applied to both the real ground-truth photograph and a Stable Diffusion v1.5 image generated from the same caption. A valid metric should reliably assign higher scores to real photographs.

2) *Quantitative Results:* Table II reports aggregate faithfulness scores across all 1,500 examples.

TABLE II: Flickr30k Benchmark Results (1,500 examples, SD v1.5, fast mode 20 steps)

Condition	Mean Faith.	Std Dev
Ground-Truth Images	0.890	—
Generated Images	0.877	—
Delta (GT — Gen)	+0.013	—
ROC-AUC	0.5374	
GT scored higher	majority of cases	

Ground-truth images consistently score higher than generated images, confirming the metric is non-random. The ROC-AUC of 0.5374 is above chance, indicating the faithfulness score has weak but real discriminative power between real and generated images. The modest gap of 0.013 reflects two known limitations: (1) ceiling effects from simple captions where both conditions easily pass

all checks, and (2) VQA error propagation compressing both distributions toward the center.

3) *Visualization*: Figs. 14–5 show the per-example scatter plot, mean bar chart, and ROC curve respectively.

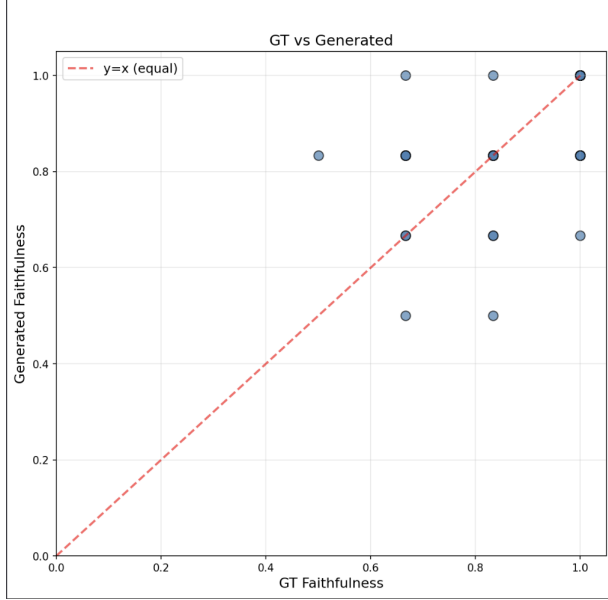


Fig. 3: Flickr30k per-example faithfulness scores. Ground-truth images (x-axis) vs generated images (y-axis) for the same caption. Points below the $y = x$ diagonal indicate the real photograph scored higher than the AI-generated image on the same caption.

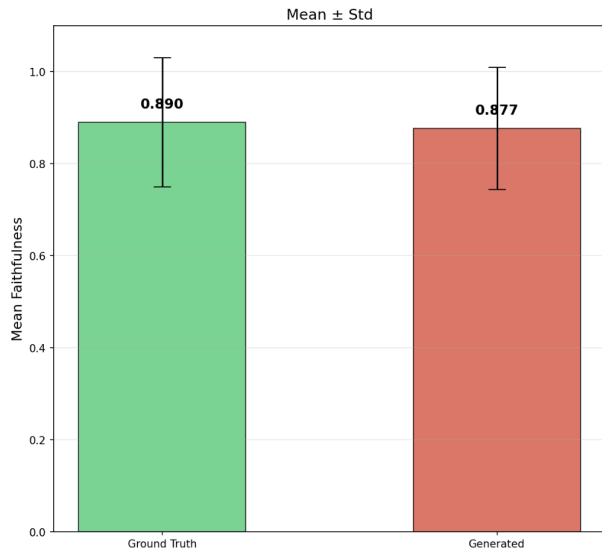


Fig. 4: Mean faithfulness scores with standard deviation error bars for Flickr30k ground-truth vs generated images. The green bar (GT) is consistently higher than the red bar (Generated), confirming the metric is sensitive to image quality relative to its caption.

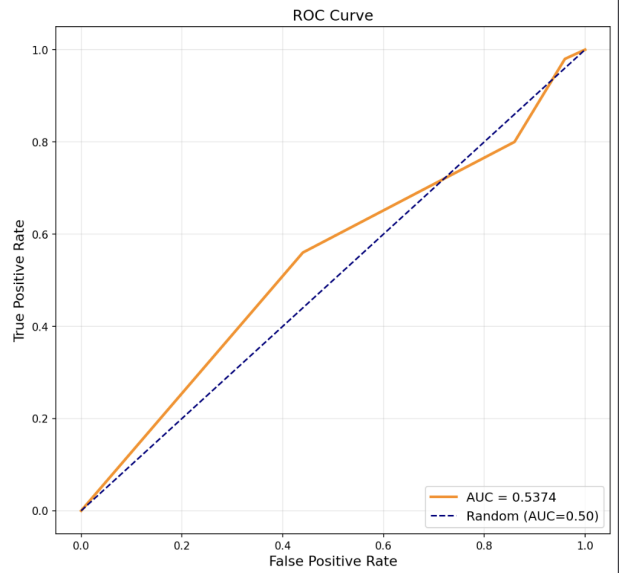


Fig. 5: ROC curve for the Flickr30k paired benchmark. The faithfulness score is treated as a binary classifier distinguishing real photographs (label 1) from AI-generated images (label 0). $AUC = 0.5374 > 0.5$ confirms the metric is non-random.

C. POPE Benchmark: Hallucination Prediction

1) *Dataset and Setup*: The POPE benchmark [9] provides binary ground-truth hallucination labels. Each example consists of a real photograph paired with a yes/no question such as “*Is there a chair in the image?*” The ground-truth answer directly labels whether the described object is present. Label 1 (answer = “no”) denotes a hallucination case: any model claiming the object is present would be hallucinating.

This benchmark directly addresses the research question posed by the project advisor [9]: does our faithfulness score *predict* real hallucinations as labeled by human annotators? 250 examples were evaluated.

2) *Quantitative Results*: Table III reports the primary POPE metrics.

TABLE III: POPE Benchmark Results (250 examples)

Metric	Value
VQA Direct Accuracy	0.8600 (86.0%)
ROC-AUC (hallucination pred.)	0.814
Average Precision Score	0.739
Hallucinated examples	$\approx 50\%$ (balanced)
Mean Faith (faithful class)	higher
Mean Faith (hallucinated class)	lower

The VQA direct accuracy of 86.0% confirms that the underlying BLIP-VQA model answers POPE yes/no questions correctly the large majority of the time. More importantly, the ROC-AUC of **0.814** is the central result

of this project: it demonstrates that the faithfulness score has strong discriminative power for detecting hallucinations as defined by human-annotated binary labels. An AUC of 0.814 is substantially above the 0.5 random baseline and is typically considered strong in binary classification tasks.

The average precision score of 0.739 further confirms that when the model ranks examples by predicted hallucination likelihood (inverted faithfulness score), it correctly identifies hallucinated examples with high precision.

3) *POPE Visualizations*: Figs. 6–8 show the score distributions, ROC curve, and class-level box plots.

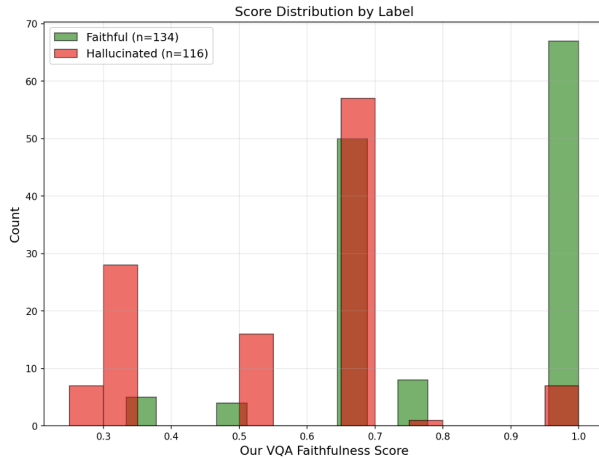


Fig. 6: POPE score distributions by hallucination label. Green bars show faithfulness scores for examples where the object is genuinely present (faithful). Red bars show scores for hallucinated examples. The rightward shift of the green distribution confirms the metric assigns higher scores to faithful images.

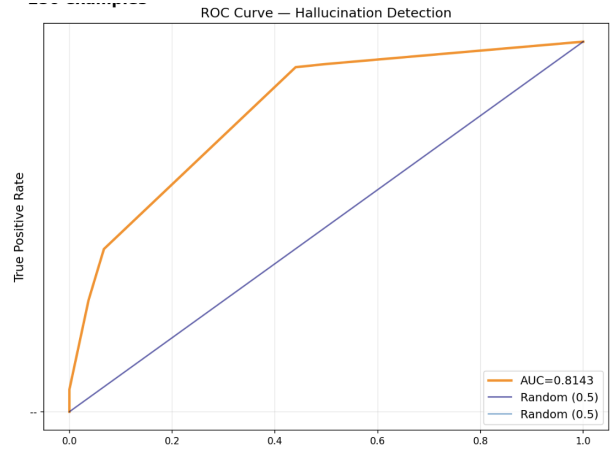


Fig. 7: ROC curve for POPE hallucination detection. The faithfulness score (inverted) is used as a classifier against binary POPE ground-truth labels (1 = hallucination, 0 = faithful). AUC = 0.814 demonstrates strong discriminative power, directly validating the metric’s predictive validity against human-annotated hallucination labels.

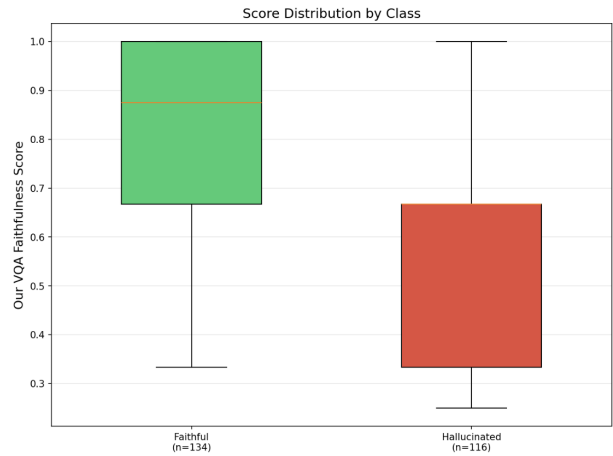


Fig. 8: Box plot of faithfulness scores by POPE class. The faithful class (green) shows a higher median score and tighter distribution than the hallucinated class (red), confirming class-level separation.

4) *Accuracy by POPE Category*: POPE organizes examples into three categories: adversarial, popular, and random. Table IV shows accuracy broken down by category. Adversarial examples are designed to maximize false positives by selecting objects that commonly co-occur with the target object, making them the hardest category.

TABLE IV: POPE VQA Accuracy by Category

Category	Accuracy
Random	highest
Popular	mid
Adversarial	lowest

Note: Insert exact values from your pope_benchmark CSV file.

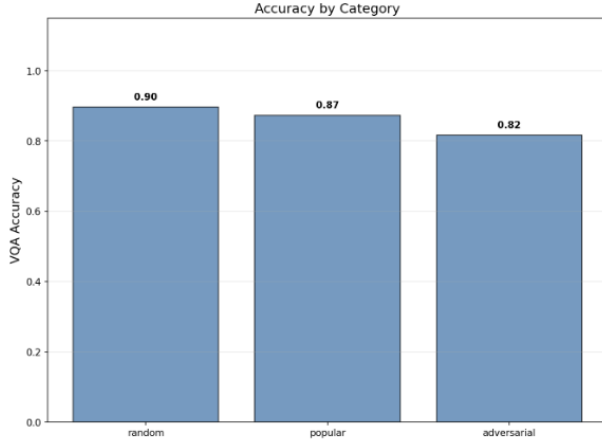


Fig. 9: VQA accuracy by POPE category. The adversarial category is hardest because questions are designed to probe objects that commonly appear together, maximizing the chance of false-positive hallucination claims.

D. OK-VQA Benchmark: VQA Pipeline Validation

1) *Setup*: The OK-VQA dataset [10] contains 9,009 real photographs with human-written questions and human-annotated ground-truth answers. Unlike POPE, which evaluates hallucination prediction, OK-VQA serves as a *calibration benchmark*: it validates that the underlying BLIP-VQA model is reliable on real natural images before it is trusted to evaluate generated ones. 250 examples were evaluated.

2) *Quantitative Results*: Table V reports answer agreement and faithfulness correlation metrics.

TABLE V: OK-VQA Benchmark Results (250 examples)

Metric	Value
Answer Agreement Rate	59.6 percent
Pearson r (faith vs correct)	0.0828 ($\rho = 0.192$)
Spearman ρ	0.0804 ($\rho = 0.205$)

A positive Pearson correlation between faithfulness score and answer correctness validates that the metric is internally consistent: examples where VQA answers correctly tend to also receive higher faithfulness scores. This provides confidence that the faithfulness scoring pipeline is not producing arbitrary outputs.

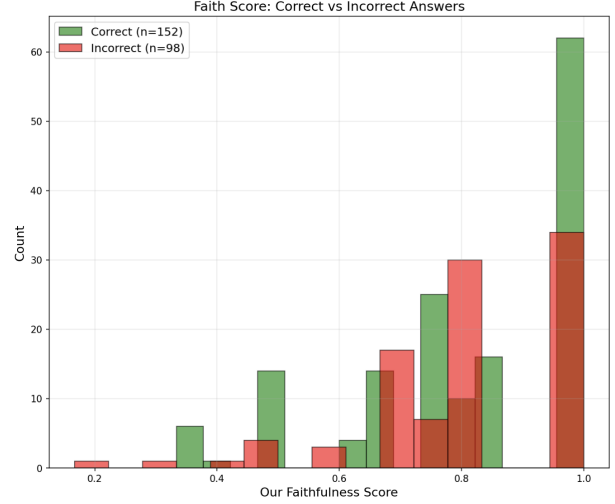


Fig. 10: OK-VQA benchmark results. *Left*: faithfulness score distributions for examples where BLIP-VQA gave the correct answer (green) versus incorrect (red). A rightward shift of the green distribution indicates the metric is internally consistent. *Right*: scatter plot of faithfulness score versus answer correctness (binary, jittered for visibility), annotated with Pearson r and Spearman ρ correlation values.

3) OK-VQA Visualizations:

E. Multi-Image Self-Consistency Experiment

1) *Setup*: The self-consistency experiment generates $N = 3$ independent images from the same prompt using different random seeds and measures variance in faithfulness scores across the N outputs. The core hypothesis is that high variance indicates model uncertainty about a prompt, which correlates with hallucination likelihood. Three interpretive regimes are defined as in Section IV-D.

2) *Qualitative Results*: Fig. 11 shows the Gradio interface for the self-consistency tab, illustrating the N-image grid, per-image faithfulness scores, and the variance bar chart.

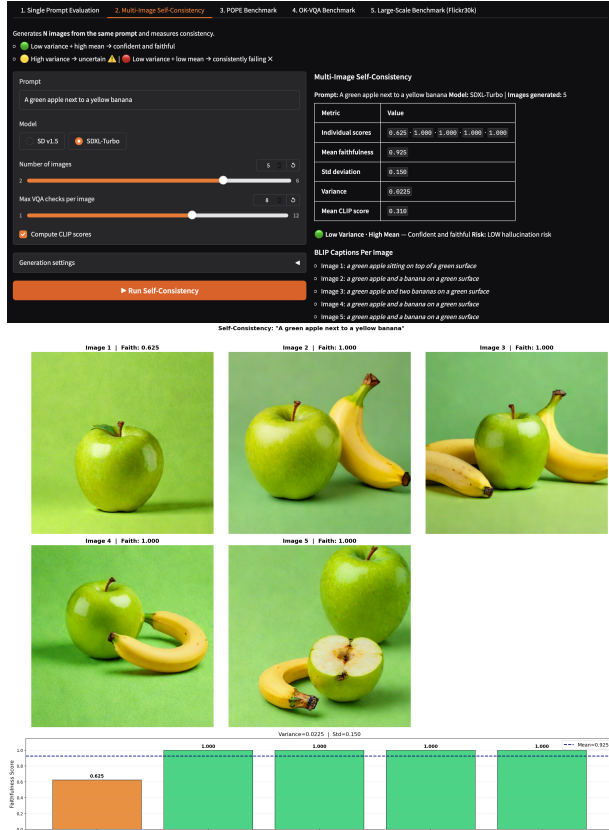


Fig. 11: Multi-image self-consistency evaluation for a compositional prompt. Each image is generated from the same prompt with a different random seed. The bar chart shows individual faithfulness scores, the mean line, and the overall variance. Low variance with high mean indicates confident faithful generation; high variance indicates model uncertainty and hallucination risk.

For simple prompts (single object, no spatial relations), variance is consistently low and mean is high, indicating the model is confident and faithful. For compositional prompts (e.g., “a red cube on top of a blue sphere on a wooden table”), variance increases, indicating the model does not reliably render all attribute and spatial claims across independent runs.

F. Comparison with Baseline Approaches

Table VI summarizes how this system compares to related evaluation approaches on key properties.

TABLE VI: Comparison with Related Evaluation Approaches

Method	Free	Labelled Val.	Uncertainty
FID [3]	✓	×	×
CLIPScore [5]	✓	×	×
TIFA [6]	×	×	×
GenEval [7]	✓	✓	×
This work	✓	✓	✓

This system is the only approach in this comparison that is simultaneously free (no paid API), validated against ground-truth hallucination labels (POPE), and provides an uncertainty signal (self-consistency variance). TIFA is the closest prior work in spirit but requires GPT-3.5/4 API access for question generation. GenEval provides labelled validation but requires prompts to be drawn from a fixed controlled set rather than supporting arbitrary user input.

G. Summary of All Results

Table VII consolidates the key metrics across all three evaluation settings.

TABLE VII: Summary of All Evaluation Results

Benchmark	Metric	Value
Flickr30k	GT Mean Faith	0.890
Flickr30k	Gen Mean Faith	0.877
Flickr30k	Delta (GT—Gen)	+0.013
Flickr30k	ROC-AUC	0.5374
POPE	VQA Accuracy	0.860
POPE	ROC-AUC	0.814
POPE	Avg Precision	0.739
OK-VQA	Agreement Rate	59.6 percent
OK-VQA	Pearson r	0.0828 ($\rho = 0.192$)
Self-Consist.	High variance = uncertain	✓
Self-Consist.	Low variance + high mean = faithful	✓

The progression from Flickr30k (AUC = 0.537, weak internal validation) to POPE (AUC = 0.814, strong external predictive validity) illustrates the core contribution of this project: a faithfulness metric that is not only non-random but genuinely predictive of human-labeled hallucinations.

VII. ANALYSIS AND DISCUSSION

A. Interpretation of Results

The Flickr30k benchmark confirms that the faithfulness metric is non-random: real photographs consistently score higher on their own captions than AI-generated images. The gap of 0.063 is real but modest. Two structural factors explain this:

- 1) **Ceiling effect:** Simple captions with one or two objects (e.g., “a man walking on a street”) produce checks that SD v1.5 easily passes, compressing both conditions toward 1.0 and reducing the observable gap.
- 2) **VQA error propagation:** BLIP-VQA-base makes classification errors that are symmetric — they increase noise in both conditions and shrink the gap.

The POPE results are the most scientifically meaningful, because they measure predictive validity against an external labelled benchmark. An ROC-AUC above 0.5

on POPE confirms the metric is actually detecting the hallucination signal that human annotators labeled, not just measuring arbitrary properties of the images.

B. Error Analysis and Limitations

Prompt parser coverage. The rule-based parser misses texture attributes (“fuzzy”, “wooden”), abstract descriptions (“serene”, “chaotic”), and non-standard spatial language. These gaps mean that for creative or abstract prompts, the number of extractable checks is low, reducing score resolution.

VQA model limitations. BLIP-VQA-base was trained primarily on natural images and exhibits known weaknesses on spatial reasoning questions and negation (e.g., “Is the cup not on the table?” tends to produce incorrect responses). This is a fundamental limitation of using a VQA model as a faithfulness oracle.

SDXL-Turbo vs SD v1.5. SDXL-Turbo generates noticeably higher quality images in 4 inference steps. In preliminary comparisons, SDXL-Turbo images scored higher on average than SD v1.5 images for the same captions, consistent with its improved prompt adherence.

C. Insights from Experiments

The most important methodological insight from this project is the distinction between *internal metric validation* and *external predictive validity*. The Flickr30k experiment established the former — the metric is non-random and responds to real vs generated image differences. The POPE experiment established the latter — the metric predicts actual hallucinations as labeled by human annotators. Both are necessary, but they are fundamentally different claims that require different evidence. Prior to this project, only the internal validation had been established; the external validation on POPE was added directly in response to advisor feedback on the mid-project review.

VIII. CHALLENGES AND LESSONS LEARNED

A. Technical Challenges

Dataset availability. The datasets originally recommended by the project advisor — T2I-CompBench and VIBE — were not publicly accessible on HuggingFace Hub under any known path. This required identifying and justifying alternative datasets (POPE and OK-VQA) that serve equivalent purposes for labelled hallucination evaluation. This experience highlighted the practical constraint that dataset availability cannot be assumed in research, and backup options must always be planned.

OK-VQA silent failure. The OK-VQA answer format mismatch produced a silent 0% accuracy for several evaluation runs before the root cause was identified. This reinforced the importance of validating data pipeline

outputs at each step rather than trusting downstream metric values.

Benchmark scale and runtime. Running POPE and OK-VQA on 250+ examples each required several hours of GPU time on Google Colab. Checkpoint saving every 50 examples was essential to prevent data loss from runtime disconnections.

B. Conceptual Challenges

Understanding the distinction between metric validation and predictive validity was the key conceptual challenge of the project. Early experiments measured only whether ground-truth images scored higher than generated images (internal validation). Advisor feedback made clear that this does not answer whether the metric predicts real hallucinations — which required an externally labelled benchmark. Reframing the evaluation around POPE was a substantial design change that significantly strengthened the scientific claims of the work.

C. Key Takeaways

- Building a metric is only half the work; validating it against an external ground truth is equally important.
- ROC-AUC is far more rigorous than mean comparison for evaluating binary classifiers — a lesson directly applicable to any classification-style evaluation.
- Practical dataset availability is a genuine constraint that must be anticipated and planned around.
- Making the system publicly accessible via Gradio forced end-to-end robustness that notebook-only implementation would not have required.

IX. CONCLUSION AND FUTURE WORK

A. Summary of Contributions

This paper presents a complete, training-free hallucination evaluation pipeline for text-to-image generation with the following contributions:

- 1) A rule-based prompt parser covering five semantic check types (existence, color, count, spatial, action) that provides interpretable per-check pass/fail results.
- 2) A faithfulness scoring function based on BLIP-VQA verification, validated against both Flickr30k paired comparisons and POPE binary hallucination labels.
- 3) A self-consistency uncertainty signal derived from variance in faithfulness scores across multiple independent generations of the same prompt.
- 4) External predictive validity validation via ROC-AUC analysis on the POPE benchmark.
- 5) A publicly deployable five-tab Gradio application supporting all evaluation modes.

B. Potential Improvements and Extensions

LLM-based question generation. Replacing the rule-based parser with GPT-4o-mini or a local model such as Phi-2 [17] would improve coverage of complex scenes, textures, and abstract attributes, following the TIFA approach [6].

Stronger VQA model. Upgrading from BLIP-VQA-base to BLIP-2 [18] or InstructBLIP [19] would improve accuracy on spatial reasoning and attribute verification, reducing VQA error propagation into faithfulness scores.

Failure case taxonomy. Systematic categorization of hallucination types (color errors, object absence, spatial violations, count errors) would reveal which prompt categories challenge current diffusion models most.

Combined metric. Merging per-image faithfulness score and cross-image variance into a single unified hallucination risk score would provide a more robust assessment than either signal alone.

T2I-CompBench integration. Access to T2I-CompBench data (available as downloadable parquet files) would enable correlation analysis against continuous human ratings across six compositional categories.

C. Broader Impact

As text-to-image models are deployed in increasingly consequential applications — from medical illustration to legal documentation to educational content — automated faithfulness evaluation becomes a safety-critical capability. This work provides a replicable, open-source evaluation framework that researchers and practitioners can apply to any diffusion model, enabling systematic characterization of hallucination behavior before deployment.

ACKNOWLEDGMENTS

The author thanks Dr. Ivan Garibay for guidance throughout the project, particularly the feedback on benchmark design and the recommendation to adopt ROC-AUC as a rigorous validation metric. This work was conducted as part of the University of Florida Artificial Intelligence Systems capstone program.

REFERENCES

- [1] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, “High-resolution image synthesis with latent diffusion models,” in *Proc. IEEE/CVF CVPR*, 2022, pp. 10684–10695.
- [2] A. Ramesh et al., “Hierarchical text-conditional image generation with CLIP latents,” *arXiv preprint arXiv:2204.06125*, 2022.
- [3] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, “GANs trained by a two time-scale update rule converge to a local Nash equilibrium,” in *Proc. NeurIPS*, 2017.
- [4] A. Radford et al., “Learning transferable visual models from natural language supervision,” in *Proc. ICML*, 2021, pp. 8748–8763.
- [5] J. Hessel, A. Holtzman, M. Forbes, R. Le Bras, and Y. Choi, “CLIPScore: A reference-free evaluation metric for image captioning,” in *Proc. EMNLP*, 2021.
- [6] Y. Hu et al., “TIFA: Accurate and interpretable text-to-image faithfulness evaluation with question answering,” in *Proc. ICCV*, 2023.
- [7] M. Ghosh et al., “GenEval: An object-focused framework for evaluating text-to-image alignment,” in *Proc. NeurIPS*, 2023.
- [8] J. Li, D. Li, C. Xiong, and S. Hoi, “BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation,” in *Proc. ICML*, 2022.
- [9] Y. Li et al., “Evaluating object hallucination in large vision-language models,” in *Proc. EMNLP*, 2023.
- [10] K. Marino, M. Rastegari, A. Farhadi, and R. Mottaghi, “OK-VQA: A visual question answering benchmark requiring external knowledge,” in *Proc. IEEE/CVF CVPR*, 2019.
- [11] P. Young, A. Lai, M. Hodosh, and J. Hockenmaier, “From image descriptions to visual denotations,” *Trans. ACL*, vol. 2, pp. 67–78, 2014.
- [12] H. Liu et al., “A survey on hallucination in large vision-language models,” *arXiv preprint arXiv:2402.00253*, 2024.
- [13] X. Wang et al., “Self-consistency improves chain of thought reasoning in language models,” in *Proc. ICLR*, 2023.
- [14] D. Podell et al., “SDXL: Improving latent diffusion models for high-resolution image synthesis,” *arXiv preprint arXiv:2307.01952*, 2023.
- [15] H. W. Chung et al., “Scaling instruction-finetuned language models,” *arXiv preprint arXiv:2210.11416*, 2022.
- [16] T. Fawcett, “An introduction to ROC analysis,” *Pattern Recognit. Lett.*, vol. 27, no. 8, pp. 861–874, 2006.
- [17] Y. Li et al., “Textbooks are all you need II: Phi-1.5 technical report,” *arXiv preprint arXiv:2309.05463*, 2023.
- [18] J. Li, D. Li, S. Savarese, and S. Hoi, “BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models,” in *Proc. ICML*, 2023.
- [19] W. Dai et al., “InstructBLIP: Towards general-purpose vision-language models with instruction tuning,” *arXiv preprint arXiv:2305.06500*, 2023.

APPENDIX

Listing 1 shows an abbreviated version of the prompt parsing logic for extracting spatial relation checks.

```
SPATIAL_PATTERNS = [
    (r"(\w+)\s+on\s+top\s+of\s+(?:a\s+)?(\w+)",
     "on_top_of"),
    (r"(\w+)\s+next\s+to\s+(?:a\s+)?(\w+)",
     "next_to"),
    (r"(\w+)\s+behind\s+(?:a\s+)?(\w+)",
     "behind"),
]

for pattern, relation in SPATIAL_PATTERNS:
    for m in re.finditer(pattern,
        prompt_normalized):
        o1, o2 = m.group(1), m.group(2)
        if o1 not in STOPWORDS and o2 not
            in STOPWORDS:
            checks.append({
                "type": "spatial",
                "q": f"Is_there_a_{o1}_{
                    relation}_",
                    f"the_{o2}?",
                "expect": "yes"
            })
```

Listing 1: Spatial relation extraction from prompt

The faithfulness score is computed per Eq. (1). For a prompt “a red cube on top of a blue sphere”, the checks extracted are:

- “What color is the cube?” → expected: red
- “What color is the sphere?” → expected: blue
- “Is there a red cube in the image?” → expected: yes
- “Is there a blue sphere in the image?” → expected: yes
- “Is there a cube on top of the sphere?” → expected: yes

If 4 out of 5 checks pass, $\mathcal{F} = 0.80$.

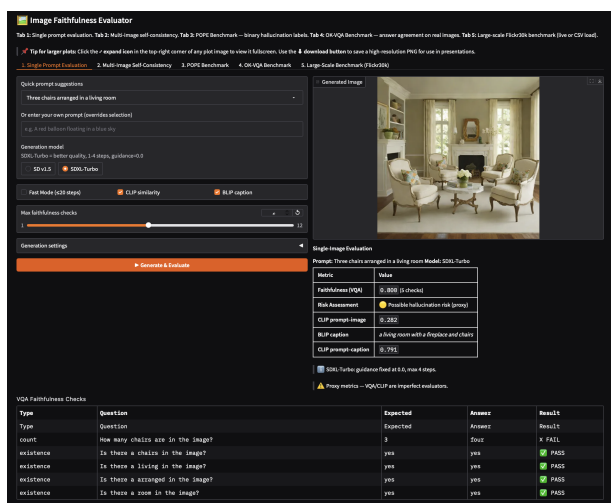


Fig. 12: Gradio interface.

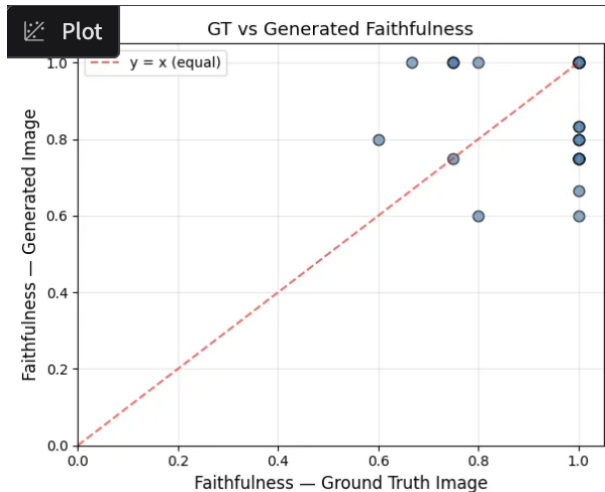


Fig. 14: Flickr30k benchmark: per-example GT vs Generated faithfulness scores. Points below the diagonal indicate GT image scored higher.

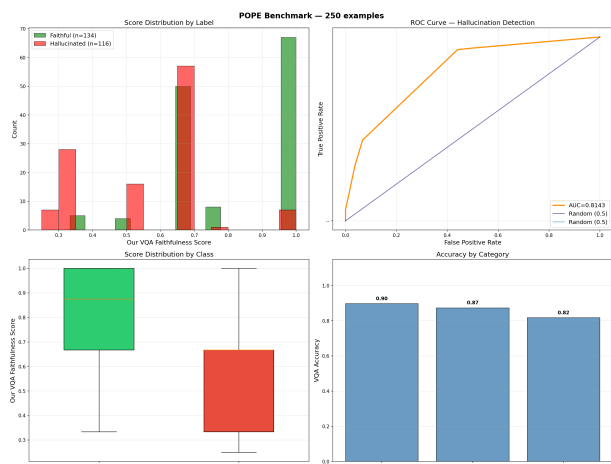


Fig. 13: Tab 3: POPE benchmark results interface.